Answers on
Booklet - 15

Taken from OCR Physics A past papers

Abbreviations, annotations and conventions used in the Mark Scheme	/	=	alternative and acceptable answers for the same marking point
	;	=	separates marking points
	NOT	=	answers which are not worthy of credit
	( )	=	words which are not essential to gain credit
	____	=	(underlining) key words which **must** be used to gain credit
	ecf	=	error carried forward

1.

(a) when two waves meet/overlap/interfere/collide/superpose (AW) B1
the resultant displacement is the sum of the displacements B1 [2]
{do not allow amplitude}
{NB allow 2 marks for good diagrams}

(b) (i) wave sources with constant phase difference B1 [1]
{NB allow "in phase" and ignore reference to frequency/wavelength/amplitude}

(ii) S_1 and S_2 'share the same light' (AW) B1

reference to diffraction at the <u>single</u> slit
OR to wavefronts e.g. "same wavefront reaches S_1 and S_2 (AW) B1 [2]

(iii) Constructive interference occurs at O B1
path difference is zero OR waves meet in phase (AW) B1 [2]

(iv) recall of formula $\lambda = ax/D$ in any valid form (e.g. $x = \lambda D/a$) C1
{NB allow undefined symbols provided they match the above as stated in the spec., otherwise they must be defined}

correct sub. with <u>consistent units</u>: $\lambda = 2 \times 10^{-3} \times 0.6 \times 10^{-3} / 1.8$ C1
$\lambda = 6.7 \times 10^{-7}$ m A1 [3]
{NB allow ecf if mm used: i.e 2 marks for 6.7×10^{-1} OR 6.7×10^{-4}}

(v) 'fringe separation' (AW) would DECREASE B1
{NB allow "more fringes would be seen"}

because $x \propto \lambda$ (AW) B1 [2]

{NB allow 'colour change' arguments for full marks:
Colour would change B1;
to a colour closer to the blue end of visible spectrum (AW) B1}

2.

<u>energy</u> of the infra-red photon is less than the <u>work function</u> of the metal surface	B1 B1	**accept** frequency and threshold frequency **or** wavelength and threshold wavelength used correctly in place of energy and work function **1 mark** only: energy of the uv photon greater than work function with no mention of ir

3.

(a)(i)		Parallel	B1
(ii)		$R = R_1R_2 / R_1 + R_2$ / $1/R_T = 1/R_1 + 1/R_2$	C1
		$R = 1.5 \times 1.0 / (1.5 + 1.0)$ / $R = 1/1.67$ / $1/R = 1.67$	C1
		$R = 0.6\ (\Omega)$	A0
(b)(i)		e.m.f. is the (total) energy (gained) / work done per (unit) charge	B1
		Energy transformed into electrical / gained by charges	B1
		OR	
		$E = W/Q$	M1
		(E = e.m.f) W = energy gained / converted to electrical and Q = charge	A1
(ii)		The chemicals (within the cell)	B1
(iii)		$R = R_1 + R_2$ / $R = 0.8 + 0.6$	C1
		$R = 1.4\ (\Omega)$	A1
(iv)		$I = 1.5 / 1.4$	C1
		$I = 1.0(7)$ (A) (Possible e.c.f)	A1
(v)	**1.**	$P = VI$ / I^2R / V^2/R	B1
	2.	$P_{int} = 1.0(7)^2 \times 0.8 = (0.916\ W)$ / $P_{ext} = 1.0(7)^2 \times 0.6 = (0.687\ W)$	C1
		ratio = 0.8 / 0.6 / ratio = 0.916/0.687	C1
		ratio = 1.3(3)	A1
		(ratio = 0.6/0.8 = 0.75 scores 2/3)	
		(ratio = 0.571, when R = 1.4Ω is used instead of 0.6Ω, scores 2/3)	
		(ratio = (1.07 × 0.8) / (1.07 × 0.6) scores 0/3)	

4.

(a)	$V = 0.014 \times 160$ $V = 2.24\ \text{V} \approx 2.2\ \text{V}$	**M1** **A0**
(b)	$V = 12 - 2.24$ $V = 9.76\ (\text{V}) \approx 9.8\ (\text{V})$	 **B1**
(c)(i)	area $= \pi \times (4.0 \times 10^{-3})^2$ / area $= \pi \times 4.0^2/10^6$ area $= 5.03 \times 10^{-5}\ (\text{m}^2) \approx 5.0 \times 10^{-5}\ (\text{m}^2)$	**M1** **A0**
(ii)	$R = \rho L/A$ (Allow any subject) $R = 1.7 \times 10^{-8} \times 0.85 / 5.0 \times 10^{-5}$ $R = 2.9 \times 10^{-4}\ (\Omega)$ (-1 if $5.0 \times 10^{-5}\ (\text{m}^2)$ is not used from c(i))	**C1** **C1** **A1**
(iii)	The resistance decreases by a factor of eight because area increases by a factor of four <u>and</u> length decreases ($3.6 \times 10^{-5}\ (\Omega)$ with correct working scores 2/2) ($3.6 \times 10^{-5}\ (\Omega)$ without working scores 1/2) (Allow 1/2 for 'Resistance decreases by a factor of four because length decreases <u>and</u> area increases by a factor of two scores')	**M1** **A1**

5.

(a)	current <u>and</u> current	B1
(b)	energy	B1
(c)(i)	(NTC) thermistor	B1
(c)(ii)	Resistance decreases when temperature is increased. (ora) (Allow correct credit for a PTC thermistor)	B1
(c)(iii)1	$I = (0.032\text{-}0.006 =)\ 0.026\ (\text{A})$	B1
(c)(iii)2	($V_{200} = 0.026 \times 200 =$) 5.2 (V) / ($V_{720} = 0.006 \times 700 =$) 4.2 (V) $E = 5.2 - 4.2$ (Allow $E = 4.2 - 5.2$) $E = 1.0$ (V) (Allow 1 sf answer) (9.4 (V) scores 1/3)	C1 C1 A1

6.

(a) ANY **2 points** made from the following:

B1 + B1 **[2]**

- reference to **nodes AND antinodes** OR **constructive AND destructive** interference
- correct link for either **antinodes with constructive** OR **nodes with destructive**
- (meeting/superposing) waves must be COHERENT (allow "in phase")

(b) two antinodes labelled (with **A**) at centres of hot zones (please look closely!) B1 **[1]**
{N.B. two or more correct scores 1, any incorrect scores zero),

(c) recall of speed of microwaves = **3 x10^8** (m/s) ---------- B1
correct substitution: e.g. λ = v/f = 3 x 10^8 / 2.45 x 10^9 ---------- C1
λ= 1.22 x 10^{-1} m = **0.122 m** ---------- A1 **[3]**

7.

(a)	**(i)** **(ii)**	node occurs where the amplitude/displacement is (always) zero antinode occurs where the amplitude (of the standing wave) takes the maximum (possible) value	B1 B1	**accept** displacement for amplitude for **(i)** only
(b)	**(i)**	wave travels to end and is reflected reflected wave interferes/superposes with incident wave always destructively at certain points to produce nodes **or** always constructively at certain points to produce antinodes	B1 B1 B1	 **accept** 2 waves of same f travelling in opposite directions interfere with no reference to reflection
	(ii)	A and N points labelled correctly	B1	
	(iii)	3	B1	
	(iv)	30 cm = λ/2 or λ = 60 cm v = f λ = 120 x 0.6 v = 72 (m s^{-1})	C1 C1 A1	**allow** 1 mark for correct calculation using v = f λ with wrong wavelength if method/reasoning clear
(c)		v = 2k becomes v = 3k (k = 36) wavelength increases by 3/2 (as frequency unchanged) 2 half wavelengths fit on the string so standing wave is set up/AW	B1 B1 B1	**accept** v increases by 3/2 or v = 108 m s^{-1} **accept** wavelength becomes 90 cm **allow ecf** correct conclusion with wrong λ

8.

i	node correctly labelled as **N** at base of cylinder antinode correctly labelled as **A** near top of cylinder	B1 B1	To score the second mark it must be the fundamental that is drawn (i.e.1/4 of a wave).
ii	0.64 = ¼ λ ⇒ λ = 4 x 0.64 = 2.56 m (or 2.6) correct substitution of cand's λ into v=f λ: e.g. f = 330/2.56 f = 129 Hz OR 130 Hz (or correct ecf answer)	C1 C1 A1	Allow e.c.f. for correct determination of λ from cand's standing wave in c(i). If they use 2.6 for λ f= 127 Hz for 2 marks Also allow ecf within this section for correct use of incorrect λ or v
i	**N** shown roughly at centre of tube and **A**s at each end	B1	Look at letters only
ii	frequency = 258 Hz OR 2 x cand's answer for c (ii)	B1	Allow e.c.f. from d (i)

9.

A correct potential divider circuit with LDR, voltmeter, (variable) resistor and cell (allow a battery) **B1**
Correct symbols for LDR, cell (allow battery), variable resistor and voltmeter **B1**

(-1 for an error or omission)

Any **five** from: (These must be stated and not from the diagram)

The (variable) resistor and LDR are connected in series (to the cell) **B1**
The voltmeter is placed across (variable) resistor / LDR **B1**

$V = \frac{R_2 V_0}{R_1 + R_2}$ / $\frac{V_1}{V_2} = \frac{R_1}{R_2}$ quoted **B1**

The resistance of LDR decreases as intensity of light increases (AW) **B1**
Correct description of how voltage/voltmeter reading changes with light intensity **B1**
The change in voltage / voltmeter reading is justified in terms of potential divider / ratio of resistance values / $I = V/(R_1+R_2)$ and $V = IR_2$ **B1**
Correct description of how the (circuit) current is affected by intensity **B1**

The variable resistor is used for 'sensitivity' / determining 'range' / 'calibration' / (monitor) different 'light levels' **B1**

One mark for QWC (Organisation) **B1**

10.

(a) (NTC) Thermistor B1
Resistance decreases (as temperature is increased) (wtte) B1

(b)(i)

$V = V_0 \times R_2 / R_1 + R_2$	$V = IR$ and $R = R_1 + R_2$ C1
$V = 9.0 \times 4.2 / (4.2 + 1.2)$	$I = 9.0 / 5400 = 1.67 \times 10^{-3}$ (A) C1
	$V = 1.67 \times 10^{-3} \times 4200$
$V = 7.0$ (V) *(Allow 7 (V)*	$V = 7.0$ (V) *(Allow 7 (V)* A1

(If R = 4200Ω or 1200 Ω used to find circuit current, then 0/3)
(Answer of 2.0 (V) scores 2/3)

(ii) Decreases (The answer must be consistent with type of thermistor in **(a)**) B1

11.

(a)	Photoelectric (effect)	B1
(b)	Charge on plate becomes (more) positive (with time)	B1
	Negative charge(s) / electrons leave the plate	B1
(c)	More / greater (rate of emission of) electrons	B1
	More / greater (rate of arrival of) photons	B1
(d)	Minimum energy needed to remove / emit / escape / release / lift electron (from metal surface)	B1

12.

a) Any five from: B1 x 5

- Photons are involved / $E = hf$
- Surface electrons are involved.
- One-to-one exchange of energy between photon and electron
- Electrons carry negative charge, therefore removal means reduction in (negative) charge (of plate).
- Electron released when photon energy > workfunction (energy).
- Electron emission related to threshold frequency.
- Energy conserved in the interaction (between photon and electron).
- Einstein's equation mentioned ($hf = ½ mv^2 + ø$).

Any two from:
More photons (in a given time).
More electrons are removed (in a given time)
The plate loses charge quicker. B1 x 2

(b)(i)	**1.** $f = 3.0 \times 10^8 / 6.3 \times 10^{-7}$		C1
	$f = 4.7(6) \times 10^{14}$		A1
	unit: hertz / Hz		B1
	2. $E = h f$		C1
	$E = 6.63 \times 10^{-34} \times 4.76 \times 10^{14} = 3.1(6) \times 10^{-19}$ (J)	(possible e.c.f)	A1
(ii)	$N = 1 \times 10^{-3} / 3.16 \times 10^{-19}$	(possible e.c.f)	C1
	$N = 3.1(7) \times 10^{15}$ (s^{-1})		A1
(iii)	Energy of photon is greater/ blue light has shorter wavelength.		B1
	Reduced (rate) of photons / fewer photons (in a given time)		B1

13.

(a)	hf:	Photon energy / quantum of energy	B1
	ϕ:	Work function (energy for the metal)	B1
	$\frac{1}{2} mv_{max}^2$	Maximum K.E / E_k of electron	B1
(b)(i)	$f = 3.0 \times 10^8 / 6.5 \times 10^{-7}$		C1
	frequency = $4.6(2) \times 10^{14} \approx 4.6 \times 10^{14}$		A1
	unit: hertz / Hz / s^{-1}		B1
(ii)	$\phi = hf$		C1
	$\phi = 6.63 \times 10^{-34} \times 4.62 \times 10^{14} = 3.0(6) \times 10^{-19}$ (J) $\approx 3.1 \times 10^{-19}$ (J) (Possible ECF)		C1
	$\phi = 3.0(6) \times 10^{-19} / 1.6 \times 10^{-19}$		C1
	work function energy = 1.9(1)(eV)		A0
	(Answer of 1.91(eV) without supporting calculations scores 2/3)		
(c)	(Rate / number of) electrons / photons increases		C1
	(Rate of / number of) photons double, ∴ (Rate of / number of) electrons double		A1

14.

a		$Q = It$ (current or charge =) 4.2×10^{-9} (number per second =) $\frac{4.2\times10^{-9}}{1.6\times10^{-19}}$ $2.6(3) \times 10^{10}$ (s^{-1})	C1 C1 C1 A0	**Allow** 3 marks for: number per second = $\frac{4.2\times10^{-9}}{1.6\times10^{-19}}$ **Allow** 2 marks for bald $2.6\underline{3} \times 10^{10}$ (s^{-1}) **Allow** 2 marks for $\frac{4.1\times10^{-9}}{1.6\times10^{-19}} (= 2.5(6)\times10^{10})$ - misread graph **Allow** 2 marks for $\frac{4.2}{1.6\times10^{-19}} (= 2.63\times10^{19})$ - POT error **Reject** bald answer of 2.6×10^{10} (s^{-1}) – since answer is given
b		$\phi = 1.6\times10^{-19} \times 2.2 (= 3.52\times10^{-19})$ $hf = \phi + KE_{(max)}$ / $\frac{hc}{\lambda} = \phi + KE_{(max)}$ KE = $(\frac{6.63\times10^{-34}\times3.0\times10^{8}}{2.5\times10^{-7}}) - 3.52\times10^{-19}$ kinetic energy = $4.4(36) \times 10^{-19}$ (J)	C1 C1 C1 A1	**Allow** 3 marks if candidate reaches as far as this third marking point – any subject (**Note**: frequency = 1.2×10^{15} (Hz)) **Allow** all 4 marks for a bald 4.4×10^{-19} (J) **Allow** 2 sf or more for the answer **Special case**: 4×10^{-19} (s^{-1}) scores 3 marks
c	i	No change (AW)	B1	
	ii	The (photoelectric) current is proportional to the intensity (AW) Increasing the intensity increases the (rate of) photons	B1 B1	These are **independent** marks For the second marking point the reference must be to the **photons** and not the electrons (since this idea has already been given a mark above)

15.

(a) Any one from: B1
Travel through vacuum / 'free space'
Travel at the speed of light / c / 3×10^8 m s^{-1} (in vacuum)
They are all transverse waves / can be polarised
Consist of oscillating electric / magnetic fields
Consist of photons

(b) gamma rays, visible light, infrared and microwaves B1

(c)(i) The minimum energy required to remove an electron (from the metal surface) B1

(c)(ii) Potassium (has the lowest threshold frequency) M1
$\phi = hf_{(o)}$ with some explanation / $f_{(o)} \propto \phi$ / threshold frequency is (directly) proportional to work function (energy) A1

(c)(iii)1. Any three from the statements 1 to 4:

1. Photon mentioned B1
2. A single photon interacts with a single electron B1
3. Energy is conserved between photon-electron interaction (wtte) B1
4. Electron is released when photon energy > / = work function (energy) / frequency > / = threshold frequency B1

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5. Electrons have a range of KE because some electrons are 'tightly held'/ are 'deep below the surface' / electrons make collisions with atoms / ions B1

(c)(iii)2. hf or $\frac{hc}{\lambda} = \phi + KE_{(max)}$ / $f = 9.38 \times 10^{14}$ (Hz) / (hf =) 6.22×10^{-19} (J) C1

$\phi = 1.6 \times 10^{-19} \times 3.7$ $(= 5.92 \times 10^{-19})$ C1

$$KE_{(max)} = \left(\frac{6.63 \times 10^{-34} \times 3.0 \times 10^{8}}{3.2 \times 10^{-7}}\right) - 5.92 \times 10^{-19}$$ C1

maximum kinetic energy = 3.0×10^{-20} (J) A0